Sri Lanka Institute of Information Technology



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Web-based Car Rental System

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Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

1. Introduction

RideLanka FleetHire is a Sri Lankan car rental company that rents vehicles to tourists, businesses, and local customers across multiple branches. Bookings, customer records, payments, and vehicle availability are currently managed through largely manual and disconnected processes, which makes it difficult to track vehicle availability, maintenance schedules, and payments in real time.

This report presents the stakeholder analysis and functional requirements gathered for the proposed Web-Based RideLanka FleetHire System. The system is intended to bring together customer management, fleet management, reservations, payments, customer support, and vehicle maintenance on a single online platform, based on structured requirement-gathering sessions conducted with simulated client representatives of RideLanka through the Project Discussion Bot.

1.1 Purpose

The purpose of this document is to identify the stakeholders of the Web-Based RideLanka FleetHire System and to record the functional requirements each stakeholder expects from the system, based on the requirement-gathering sessions conducted.

1.2 Scope

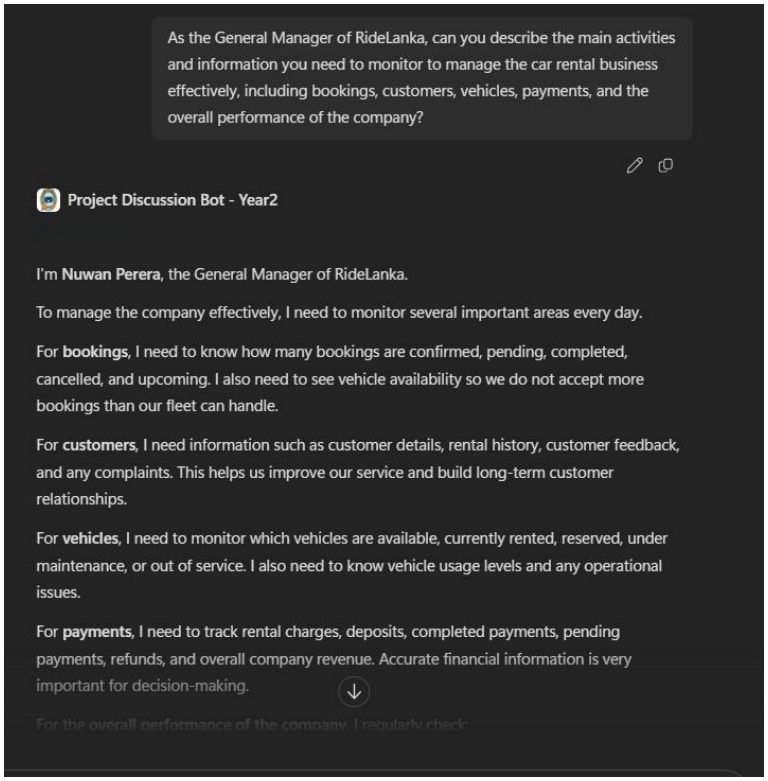
This report covers stakeholder identification and role-based functional requirements only. System design, architecture, and use-case modelling fall outside the scope of this document and will be addressed in later stages of the project.

2. Summary of Requirement Gathering Process

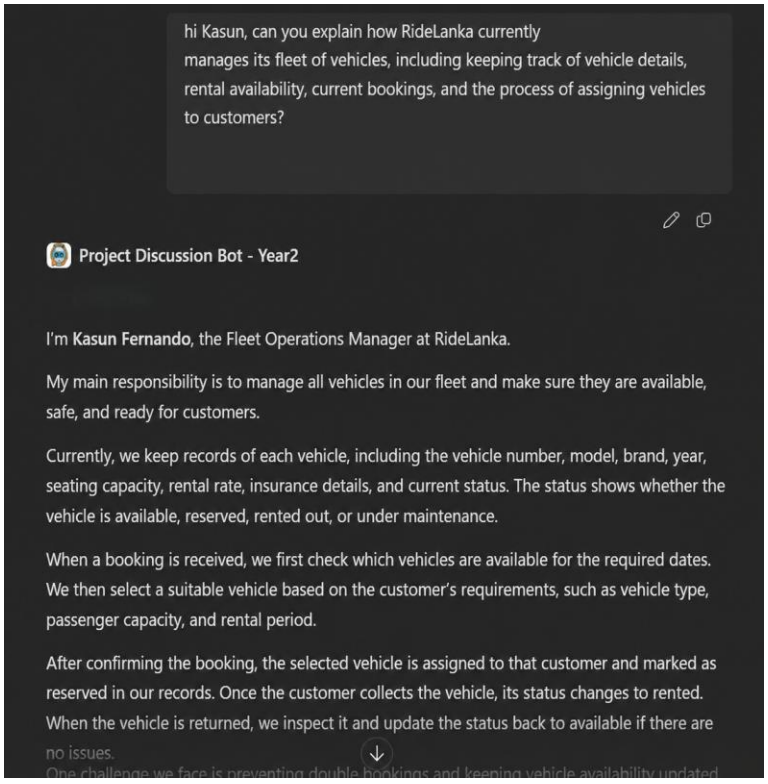
The group used the Project Discussion Bot to simulate interviews with six stakeholder personas relevant to the RideLanka FleetHire System. For each persona, the bot was asked targeted questions about that stakeholder's goals, day-to-day workflows, and the functionality they need from the system. The questions and the bot's responses were screenshotted as evidence and are used below, together with team discussion, as the basis for the functional requirements documented in Sections 4 and 5.

A minimum of one stakeholder interaction was captured per functional area, giving six pieces of evidence in total, one per group member and one per core function identified for the system.

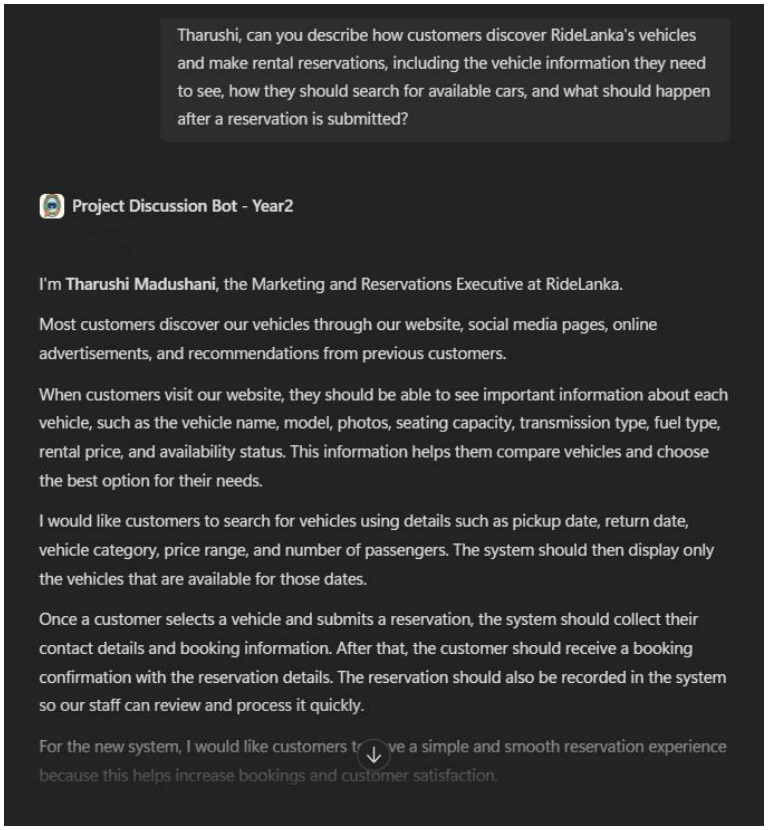
User & Customer Management — Persona: General Manager (Nuwan Perera)



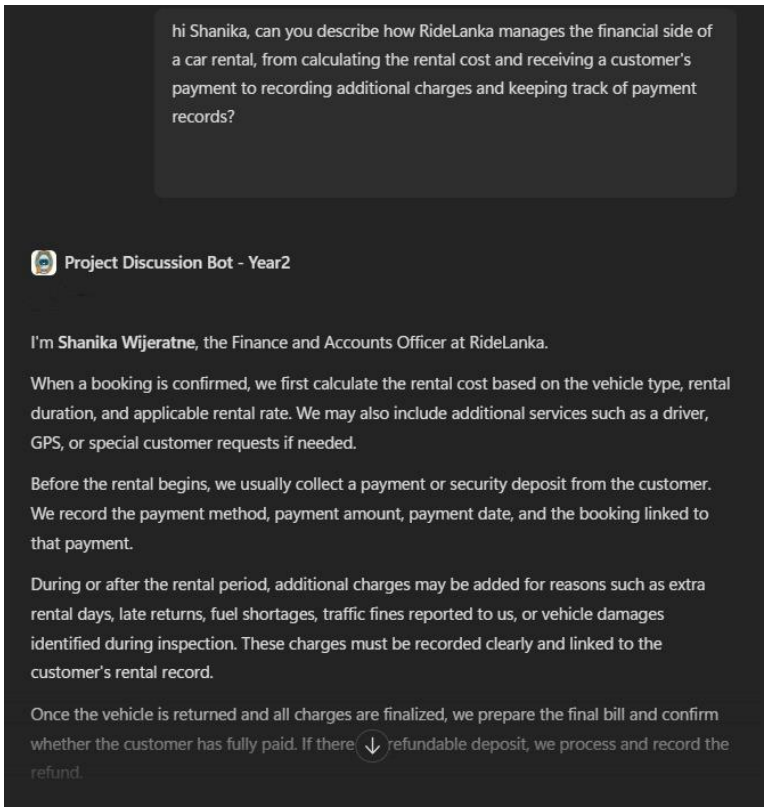
Vehicle Management — Persona: Fleet Operations Manager (Kasun Fernando)



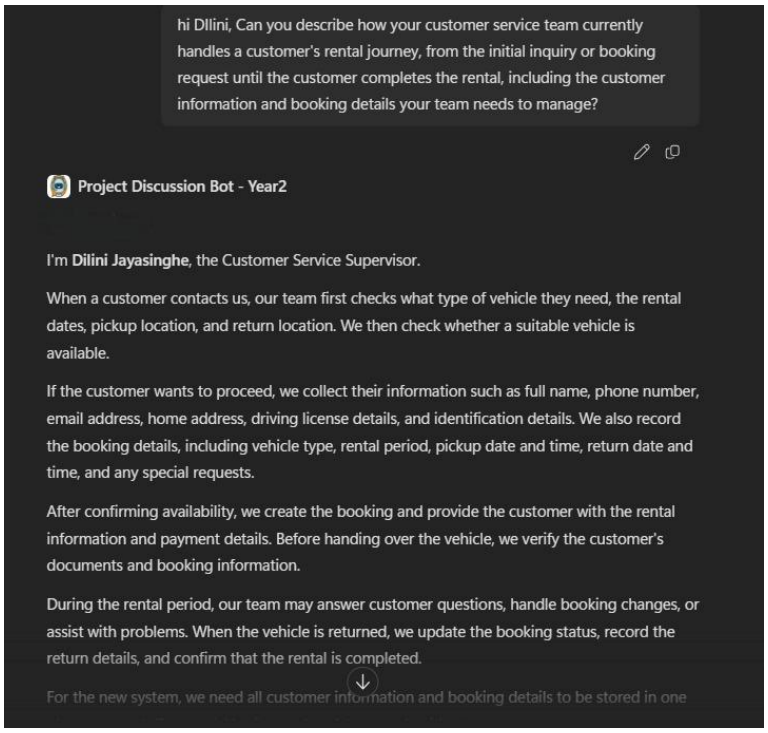
Booking & Reservation Management — Persona: Marketing and Reservations Executive (Tharushi Madushani)



Payment Management — Persona: Finance and Accounts Officer (Shanika Wijeratne)

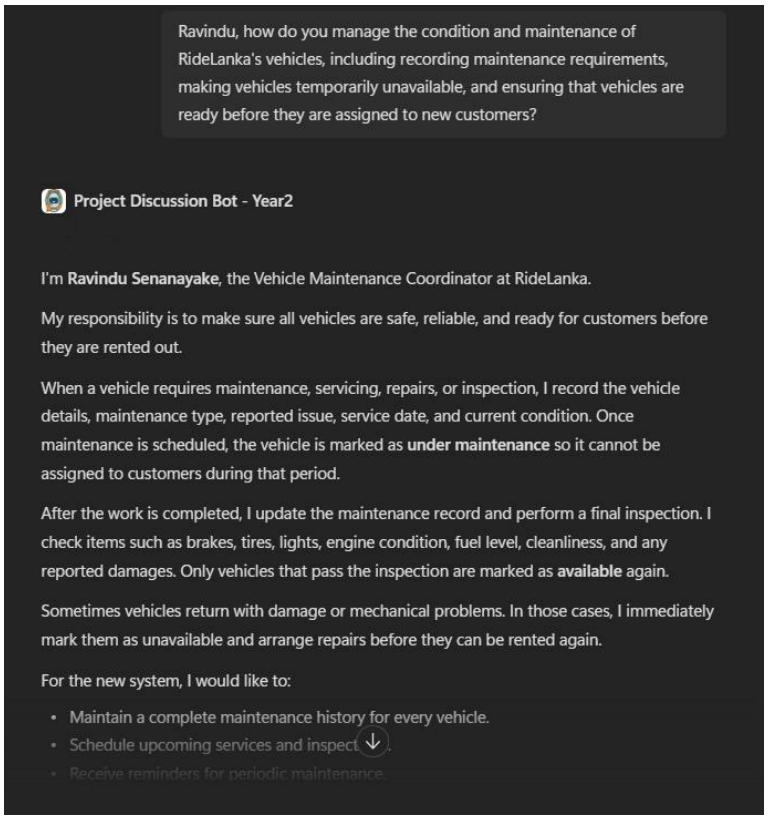


Feedback & Support Management — Persona: Customer Service Supervisor (Dilini Jayasinghe)



Screenshot: bot conversation used to elicit requirements for this stakeholder.

Vehicle Maintenance & Service Management — Persona: Vehicle Maintenance Coordinator (Ravindu Senanayake)



Screenshot: bot conversation used to elicit requirements for this stakeholder.

3. Identified Users and Their Roles

Two groups of stakeholders were identified for this project: the client representatives at RideLanka who provided the requirements, and the system end-users (actors) who will operate the finished system.

3.1 Client Representatives Consulted

Representative	Role at RideLanka	Interest in the System
Nuwan Perera	General Manager	Oversees overall business operations; needs full visibility and control over user accounts and access.
Kasun Fernando	Fleet Operations Manager	Responsible for keeping the fleet organised, available, and correctly categorised across all branches.
Tharushi Madushani	Marketing and Reservations Executive	Manages the reservation pipeline and promotional campaigns that drive bookings.
Shanika Wijeratne	Finance and Accounts Officer	Handles billing, collections, and financial reporting for every rental transaction.
Dilini Jayasinghe	Customer Service Supervisor	Owns the post-rental customer experience, including complaints and satisfaction tracking.
Ravindu Senanayake	Vehicle Maintenance Coordinator	Keeps every vehicle roadworthy and safe by coordinating servicing, repairs, and inspections.

3.2 System End-Users (Actors)

System Role	Description
Customer	Registers, searches vehicles, makes and manages bookings, makes payments, and submits feedback via the website.
Branch / Front-Desk Staff	Handles walk-in bookings, vehicle hand-over and return, and day-to-day customer service at a branch.
Fleet / Maintenance Staff	Updates vehicle status, logs servicing and repair activity, and records inspection results.
Finance Staff	Processes payments, issues invoices/receipts, and generates financial reports.
System Administrator	Manages user accounts, roles, and access permissions across the platform.

4. Final List of Core Functions

Based on the requirement-gathering sessions, six core functions were identified for the Web-Based RideLanka FleetHire System, one per group member:

#	Core Function	Documented By	IT Number
1	User & Customer Management	Rambukpitiya S. M. H. C. B. D.	IT 25 1037 36
2	Vehicle Management	Jayawardana I. K. S. P.	IT 25 1037 32
3	Booking & Reservation Management	Weerasingha W.M.K.K.G.N.A.	IT 25 1037 37
4	Payment Management	Bhagyamathi R. M. M. D.	IT 25 1037 33
5	Feedback & Support Management	Rajapaksha R. M. R. D.	IT 25 1037 34
6	Fleet Management	Dananjaya K. A. K.	IT 25 1037 35

5. Description of Each Function (with Sub-Functions)

5.1 User & Customer Management

Client Representative: Nuwan Perera — General Manager

This function gives the General Manager and administrators full visibility and control over who can access the system and what they can do, while allowing customers to manage their own profiles.

- **Account registration & login:** Customers, staff, and admins register and log in securely with role-based access control.
- **User administration:** Admins can view, add, edit, suspend, or delete user accounts across all branches.
- **Role & permission management:** Admins assign roles (customer, branch staff, finance, maintenance, admin) and control what each role can access.
- **Customer profile management:** Customers maintain contact details, driving-licence/ID verification, and view their booking history.
- **Activity log:** The system records key account actions for accountability and audit purposes.

5.2 Vehicle Management

Client Representative: Kasun Fernando — Fleet Operations Manager

This function keeps the fleet organised, correctly categorised, and visible across all branches so staff always know what is available.

- **Vehicle registration:** Add, update, or remove vehicles with details such as make, model, category, plate number, and home branch.
- **Availability tracking:** Show real-time vehicle status (available, booked, under maintenance) per branch.
- **Categorisation:** Group vehicles by type (economy, SUV, luxury, van) and rental rate.

- **Branch/location transfer:** Record movement of vehicles between branches.
- **Documentation management:** Store and track vehicle registration, insurance, and permit expiry dates.

5.3 Booking & Reservation Management

Client Representative: Tharushi Madushani — Marketing and Reservations Executive

This function manages the end-to-end reservation pipeline and supports the promotional campaigns that drive bookings.

- **Online booking:** Customers search vehicles by category, location, and date, and create a reservation.
- **Real-time availability check:** The system checks vehicle availability before confirming a booking to prevent double-booking.
- **Booking modification & cancellation:** Customers and staff can amend or cancel an existing reservation within policy rules.
- **Promotions & discount codes:** Apply marketing promotions or discount codes at checkout.
- **Reservation calendar:** Staff view a pipeline/calendar of upcoming pick-ups and returns per branch.

5.4 Payment Management

Client Representative: Shanika Wijeratne — Finance and Accounts Officer

This function handles billing, collections, and financial reporting for every rental transaction.

- **Online payment processing:** Accept card and bank-transfer payments for bookings and extra charges.
- **Invoicing & receipts:** Automatically generate invoices and receipts for each transaction.
- **Deposits & refunds:** Manage security deposits and process refunds where applicable.
- **Additional charge calculation:** Calculate late-return fees, fuel charges, and damage charges.
- **Financial reporting:** Produce revenue reports by branch, vehicle category, and time period.

5.5 Feedback & Support Management

Client Representative: Dilini Jayasinghe — Customer Service Supervisor

This function owns the post-rental customer experience, including complaint handling and satisfaction tracking.

- **Complaint submission:** Customers submit complaints or support requests linked to a specific booking.
- **Feedback & ratings:** Customers rate their rental experience after a booking is completed.
- **Ticket tracking:** Support staff assign, track, and resolve customer tickets to closure.
- **Satisfaction reporting:** Generate summary reports of customer satisfaction and recurring issues.

5.6 Vehicle Maintenance & Service Management

Client Representative: Ravindu Senanayake — Vehicle Maintenance Coordinator

This function was added to the original set of functions because none of the other areas adequately covered the fleet's servicing and roadworthiness needs, which are essential to running a safe and reliable rental operation.

- **Maintenance scheduling:** Plan routine servicing and safety inspections for each vehicle.
- **Repair logging:** Record reported faults and repair work carried out.
- **Service history:** Maintain a full service history per vehicle.
- **Maintenance alerts:** Notify coordinators of upcoming or overdue maintenance.
- **Automatic status update:** Mark a vehicle unavailable for booking while it is under service.

6. Cross-Cutting Non-Functional Requirements

The following requirements were raised consistently across multiple stakeholders and apply system-wide rather than to a single function:

#	Requirement
1	All users must log in securely before accessing role-specific functionality, with access restricted by role (customer, staff, admin).
2	Customer, payment, and vehicle information must be kept secure and confidential at all times.
3	The system must support timely notifications and alerts (booking confirmations, payment receipts, maintenance due dates, overdue returns).
4	The system must support reporting and analytics across all modules for management decision-making.
5	The system must be accessible on both desktop and mobile browsers for staff and customers.
6	The system must prevent double-booking by checking vehicle availability in real time before confirming a reservation.

7. Conclusion

This report identified six core functions for the Web-Based RideLanka FleetHire System — User & Customer Management, Vehicle Management, Booking & Reservation Management, Payment Management, Feedback & Support Management, and Vehicle Maintenance & Service Management — and documented the functional requirements gathered for each area from the corresponding stakeholder persona. Vehicle Maintenance & Service Management was added to the original set of functions because none of the other areas adequately covered the fleet's servicing and roadworthiness needs, which are essential to running a safe and reliable rental operation. These requirements form the basis for the next stages of the project, including use-case modelling and system design.